CS2001 Data Structures Project

Intelligent Forest Advisory And Multi-Structure Decision System

Project Report

Group Members

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1. Introduction

The Intelligent Forest Advisory & Multi-Structure Decision System (IFAMDS) is a C++ console-based project that simulates a real-world forest monitoring and management system. Data is processed through multiple connected data structures to support decision making.

Main features of the system include:

- Interactive menu-driven interface
- Multiple data structures working together
- Simulation of five forest emergency scenarios
- Modular and structured C++ code
- Time complexity analysis for major operations

System overview — Eight Departments

Dept.	Name	Primary Role
1	Environmental Data Acquisition	Collect & validate raw sensor data from forest zones
2	Event Memory & Temporal Reconstruction	Store time-stamped event records; enable rollback
3	Execution Control & Computational Reasoning	Step-by-step processing with backward correction and state synchronization
4	Task Scheduling & Priority Allocation	Order tasks by urgency; balance system load
5	Hierarchical Decision Intelligence	Make decisions at zone, regional, and global level
6	Spatial Connectivity & Graph-Based Routing	Model forest as a graph; compute fire-spread paths
7	Indexing & Retrieval Acceleration	O(1) key-value access with collision handling & cache
8	System Monitoring & Adaptive Optimization	Track latency, load, and bottlenecks in real time

2. Data Structures Used

The table below maps each data structure family to its instances, functional role, and the time complexity of key operations.

2.1 Arrays — Environmental Processing Layer

Instance	Type	Purpose	Key Operation	Complexit y
A1	Static 1-D	Stores fixed normal forest values as baseline	Comparison lookup	O(1)
A2	Dynamic 1-D	Live sensor stream (temperature, smoke, wind)	Append / update	O(1)
A3	Static 2-D	Forest grid spatial map	Zone retrieval [r][c]	O(1)
A4	Dynamic 2-D	Terrain expansion with live updates	Spatial interpolation	O(1) per cell

Operations performed:

- Sensor data input, spatial interpolation (average of 4 neighboring zones), boundary detection (sharp temperature changes), and anomaly filtering (temp > 45°C, smoke > 70, humidity < 20%).

2.2 Linked Lists — Event Memory Layer

Instances	Type	Purpose	Complexity
L1–L3	Singly Linked	Raw, verified & anomaly event streams (forward traversal)	O(1) insert, O(n) search
L4–L6	Doubly Linked	Forward/backward correction chains; state synchronisation	O(1) insert/delete, O(n) traverse
L7–L10	Circular	Continuous local, system-wide, emergency & stability loops	O(1) insert, O(n) full scan

Operations performed: Each node stores: (value, timestamp, zoneID). Noise is rejected when $|V_i - V_{i-1}| >= \delta$. Anomalies are flagged when $|V_{current} - V_{normal}| > \theta$. Circular lists keep monitoring running in a loop without restarting.

2.3 Queues — Scheduling Engine

Instance	Type	Purpose	Complexity
Q1	FIFO Queue	Routine monitoring tasks (minute-interval sensor updates)	O(1) enqueue/dequeue
Q2	FIFO Queue	High-frequency surveillance in sensitive zones	O(1) enqueue/dequeue
Q3	Priority Queue	Emergency response — fire/smoke alerts processed first	O(log n) insert
Q4	Multi-Factor Queue	Combined condition tasks requiring joint evaluation	O(log n) insert

Operations performed: Enqueue (add new task), Dequeue (process task), Priority Switching (emergency tasks skip ahead), Load Balancing (move tasks from Q2 to Q1 when overloaded).

2.4 Trees — Decision Intelligence Layer

Twelve tree instances are organised into four functional groups:

Group	Trees	Purpose	Decision Formula
Structural	T1, T2, T3	Zone hierarchy, sub-zone decomposition, terrain risk classification	Terrain Risk = (Slope + Dryness + Density) / 3
Resource	T4, T5, T6	Water availability, fire-control tools, equipment allocation	Priority = Risk × Impact
Incident	T7, T8, T9	Fire level, wildlife activity, human intrusion detection	Fire Level = α(Temp) + β(Smoke)
Decision	T10, T11, T12	Local, regional escalation, and global emergency decisions	Score = w1(Fire) + w2(Smoke) + w3(Temp)

Operations performed: Decision logic: if Score > 0.6, emergency is activated. Regional escalation triggers when Fire Spread Rate > 0.5. Global alert fires when ΣRiskzones > threshold.

2.5 Graphs — Spatial Routing Layer

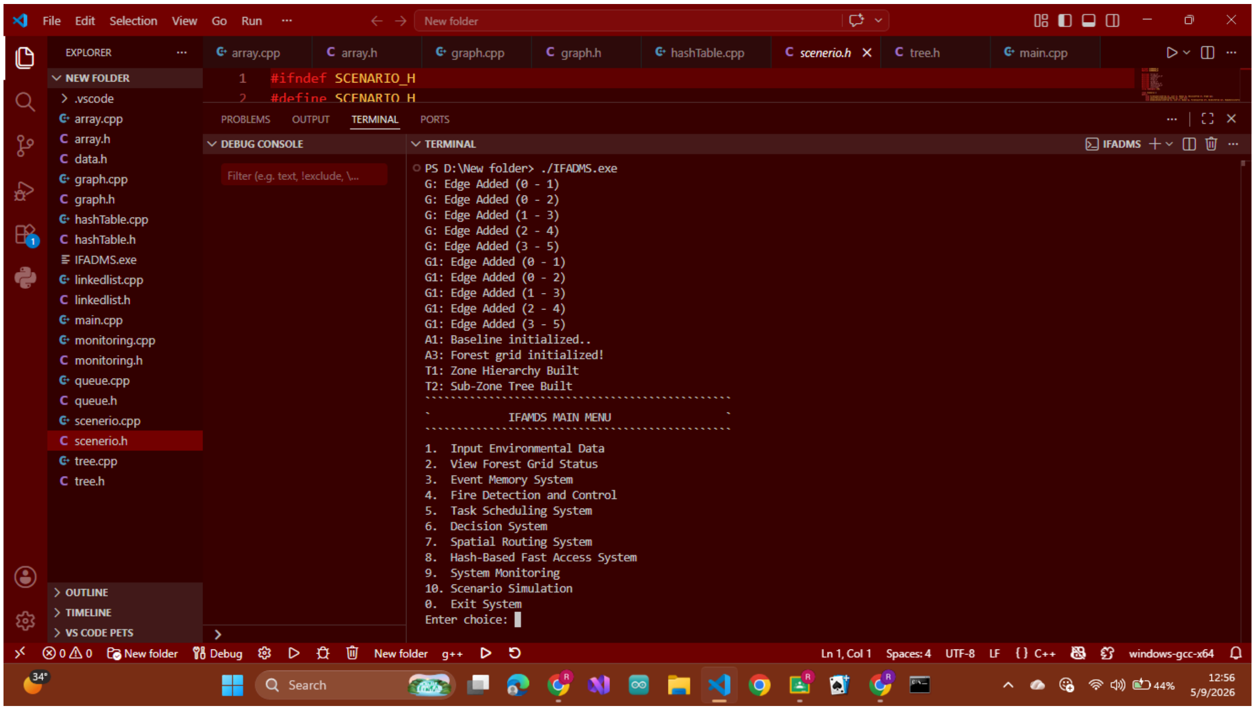
Instance	Representation	Use Case	Traversal / Cost
G1	Adjacency List	Sparse mountain/trail forest paths	BFS: O(V+E) | DFS: O(V+E)
G2	Adjacency Matrix	Dense grid monitoring across all zones	Path lookup: O(1)

Path Cost = Distance + Danger. Fire-aware update: Cost = Distance × (1 + FireLevel). BFS models level-by-level fire spread; DFS explores a single spread direction deeply. Routes are blocked dynamically as fire increases.

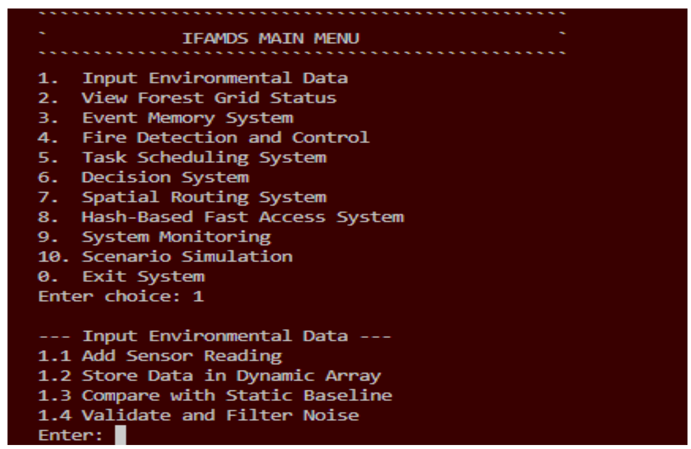
2.6 Hash Tables — Indexing & Retrieval Layer

Instance	Purpose	Mechanism	Complexity
H1 – Primary Index	ZoneID → {Temp, Smoke, Humidity}	Hash function: Index = Key mod TableSize	O(1) average
H2 – Collision Table	Handles same-index conflicts (e.g. Zone 3 & Zone 13)	Separate chaining / open addressing	O(1) avg, O(n) worst
H3 – Fast Cache	Frequently accessed zone data (e.g. Zone 1 during fire)	LRU-style retrieval cache	O(1) cache hit

3. System Modules



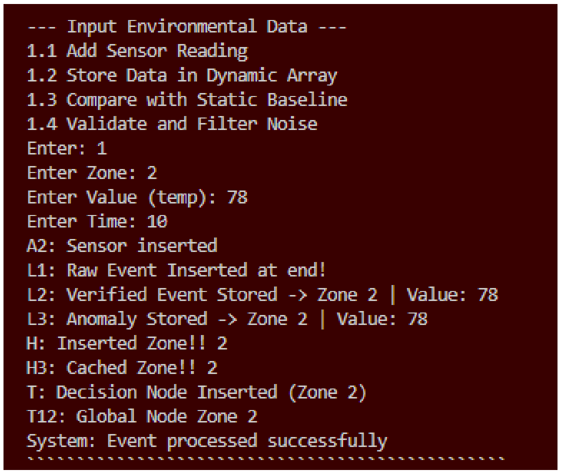
3.1 Input Environmental Data

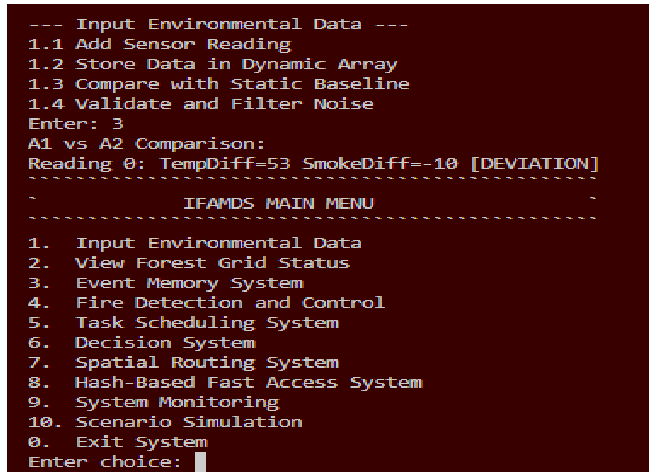


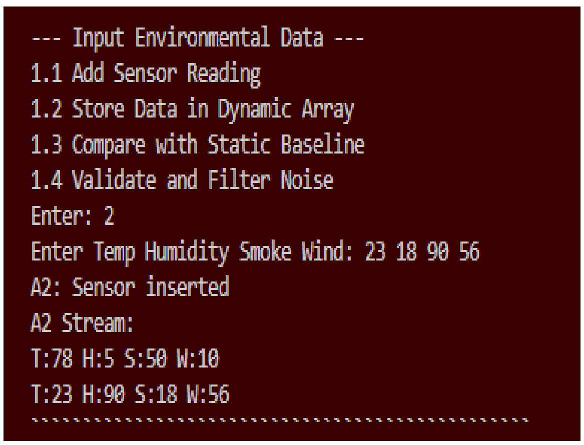
This module accepts temperature, humidity, smoke, and wind values from sensors.

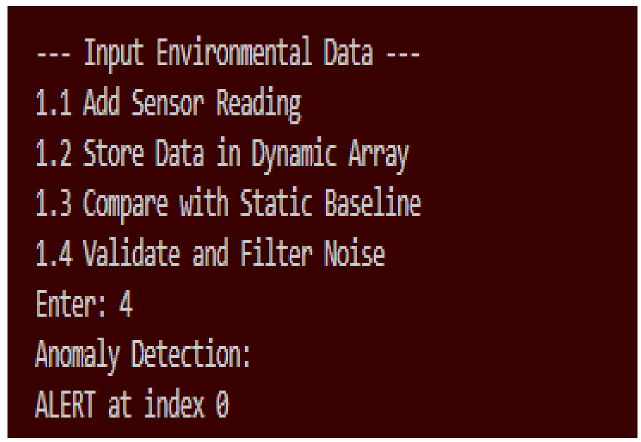
Functions:

- Add sensor readings
- Store dynamic data
- Compare with baseline
- Filter noise







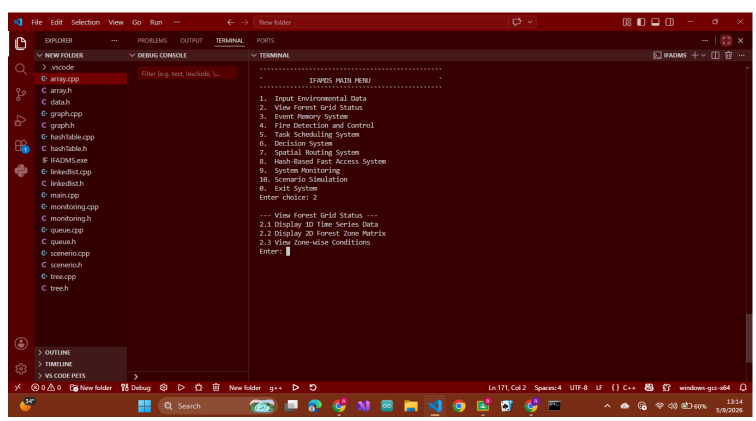


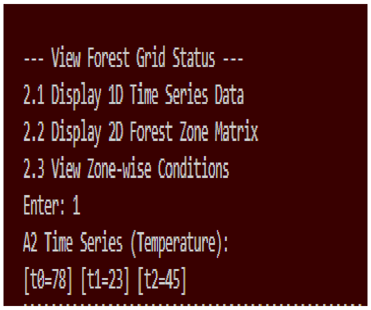
3.2 Forest Grid Status

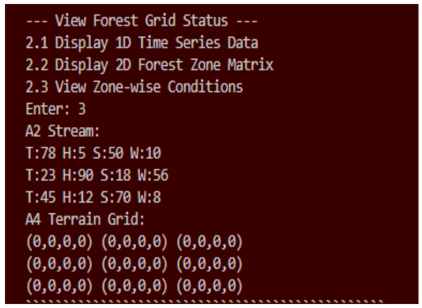
This module displays forest zone information using 1D and 2D arrays.

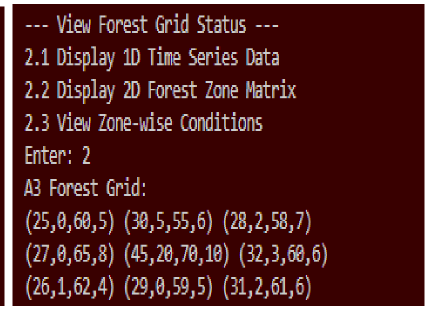
Functions:

- Show time series data
- Show forest grid
- Display zone conditions







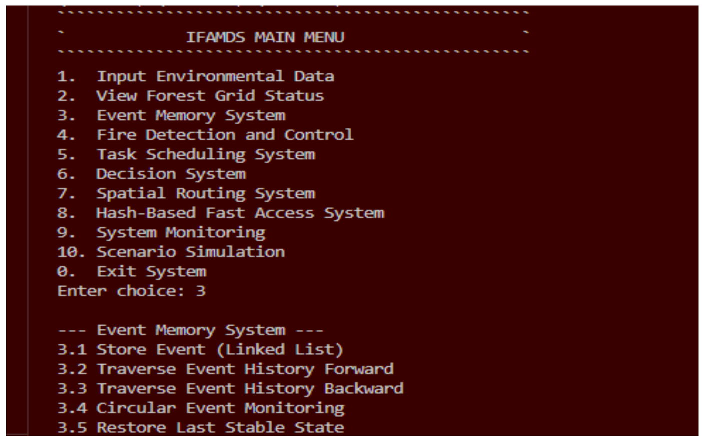


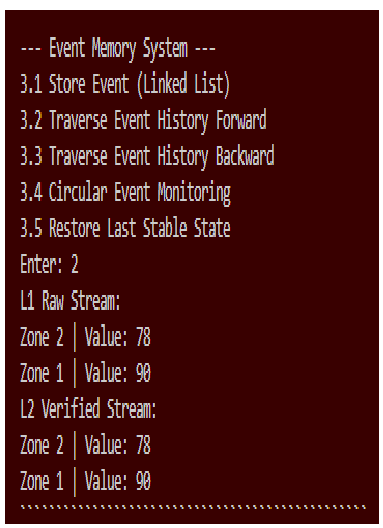
3.3 Event Memory System

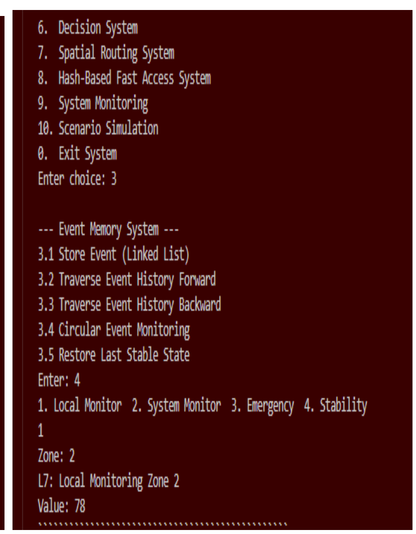
This module stores environmental events using linked lists.

Functions:

- Store events
- Traverse event history
- Circular monitoring
- Restore stable state







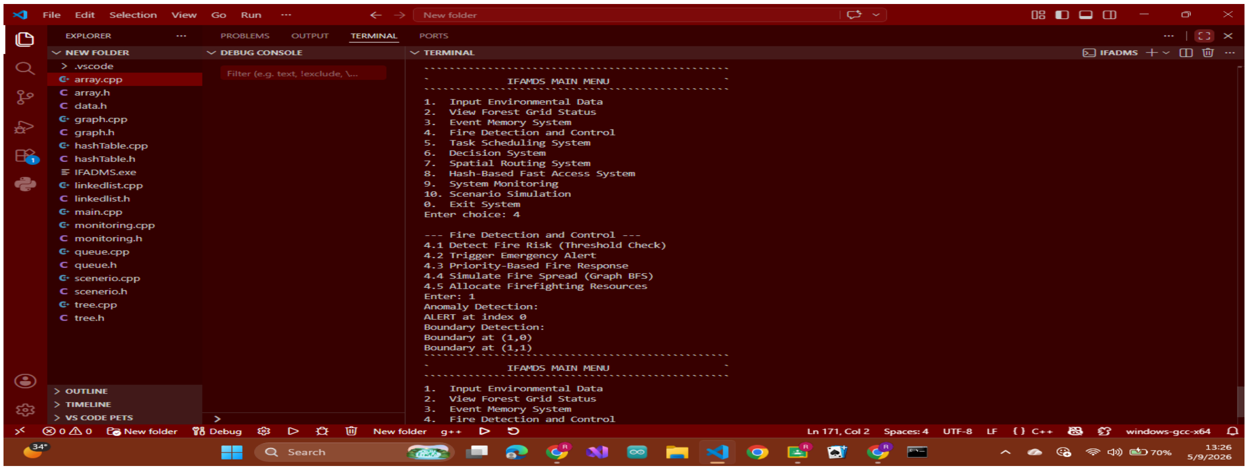
3.4 Fire Detection and Control

This module detects fire risks and manages emergency response.

Functions:

- Threshold checking

- Emergency alerts
- Fire spread simulation
- Resource allocation



```
IFAMDS MAIN MENU

1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 4

--- Fire Detection and Control ---
4.1 Detect Fire Risk (Threshold Check)
4.2 Trigger Emergency Alert
4.3 Priority-Based Fire Response
4.4 Simulate Fire Spread (Graph BFS)
4.5 Allocate Firefighting Resources
Enter: 2
Zone Value Time: 3 95 20
Q3: EMERGENCY ADDED
Q3 Processing:
Q3: ALERT Zone 3
```

```
IFAMDS MAIN MENU

1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 4

--- Fire Detection and Control ---
4.1 Detect Fire Risk (Threshold Check)
4.2 Trigger Emergency Alert
4.3 Priority-Based Fire Response
4.4 Simulate Fire Spread (Graph BFS)
4.5 Allocate Firefighting Resources
Enter: 3
Zone Value Time: 2 99 25
PROMOTED TO EMERGENCY QUEUE
Q3 Processing:
Q3: ALERT Zone 2
```

```
IFAMDS MAIN MENU

1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 4

--- Fire Detection and Control ---
4.1 Detect Fire Risk (Threshold Check)
4.2 Trigger Emergency Alert
4.3 Priority-Based Fire Response
4.4 Simulate Fire Spread (Graph BFS)
4.5 Allocate Firefighting Resources
Enter: 4
Enter start zone (0-5): 0
G BFS: 0 1 2 3 4 5
G DFS: 0 1 3 5 2 4
```

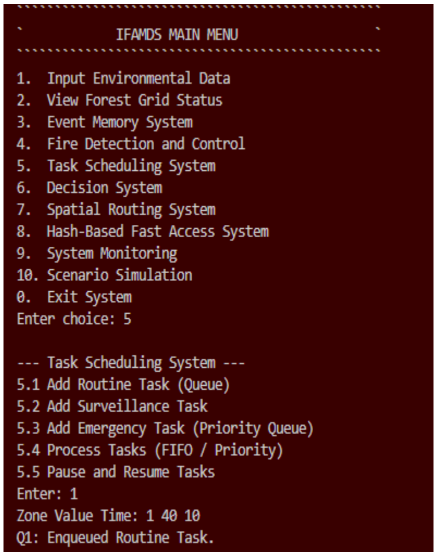
```
IFAMDS MAIN MENU

1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 4

--- Fire Detection and Control ---
4.1 Detect Fire Risk (Threshold Check)
4.2 Trigger Emergency Alert
4.3 Priority-Based Fire Response
4.4 Simulate Fire Spread (Graph BFS)
4.5 Allocate Firefighting Resources
Enter: 5
Zone ResourceLevel: 2 0.8
T5: FireControl Zone 2 Resources=0.8
T6: Equipment Zone 2 Priority=0.64
T5: Fire Control Resources:
Zone 2 Resources=0.8 [ADEQUATE]
T6: Equipment Allocation:
Zone 2 Priority=0.64 [ALLOCATE NOW]
```

3.5 Task Scheduling System

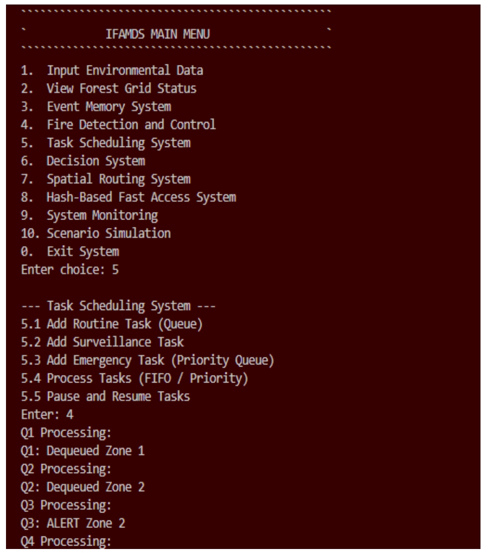
This module manages routine and emergency tasks using queues.

Functions:

- Add tasks
- Process tasks
- Pause/resume queues



```
IFAMDS MAIN MENU
1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 5

--- Task Scheduling System ---
5.1 Add Routine Task (Queue)
5.2 Add Surveillance Task
5.3 Add Emergency Task (Priority Queue)
5.4 Process Tasks (FIFO / Priority)
5.5 Pause and Resume Tasks
Enter: 1
Zone Value Time: 1 40 10
Q1: Enqueued Routine Task.
```



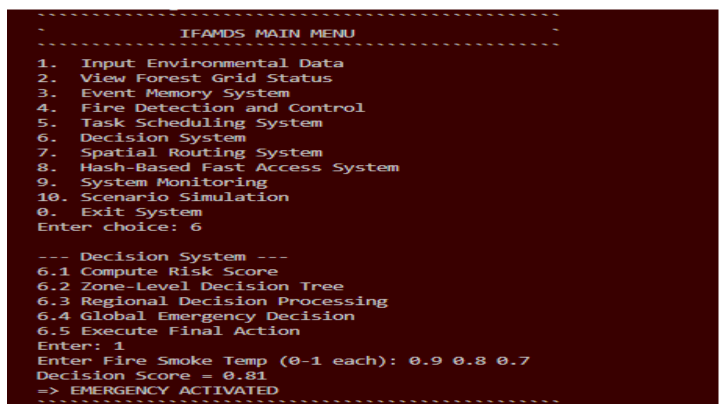
```
IFAMDS MAIN MENU
1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 5

--- Task Scheduling System ---
5.1 Add Routine Task (Queue)
5.2 Add Surveillance Task
5.3 Add Emergency Task (Priority Queue)
5.4 Process Tasks (FIFO / Priority)
5.5 Pause and Resume Tasks
Enter: 4
Q1 Processing:
Q1: Dequeued Zone 1
Q2 Processing:
Q2: Dequeued Zone 2
Q3 Processing:
Q3: ALERT Zone 2
Q4 Processing:
```

3.6 Decision System

This module calculates risk scores and makes emergency decisions.

Functions:

- Risk calculation
- Zone decisions
- Global emergency alerts



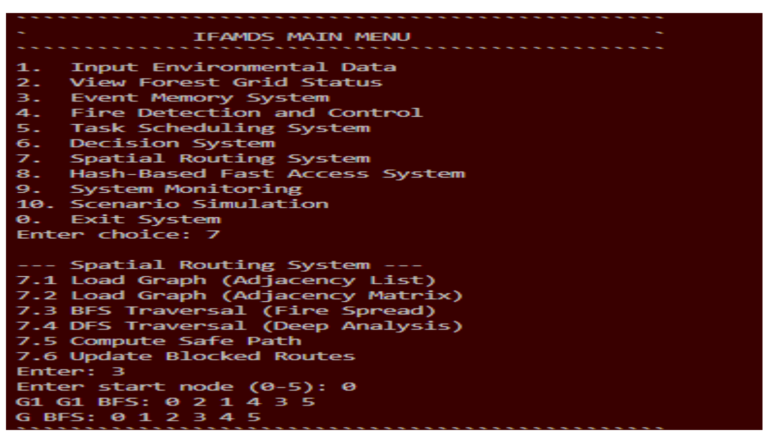
```
IFAMDS MAIN MENU
1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 6

--- Decision System ---
6.1 Compute Risk Score
6.2 Zone-Level Decision Tree
6.3 Regional Decision Processing
6.4 Global Emergency Decision
6.5 Execute Final Action
Enter: 1
Enter Fire Smoke Temp (0-1 each): 0.9 0.8 0.7
Decision Score = 0.81
=> EMERGENCY ACTIVATED
```

3.7 Spatial Routing System

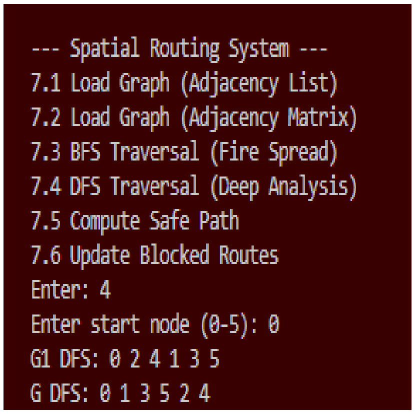
This module uses graphs for routing and traversal.

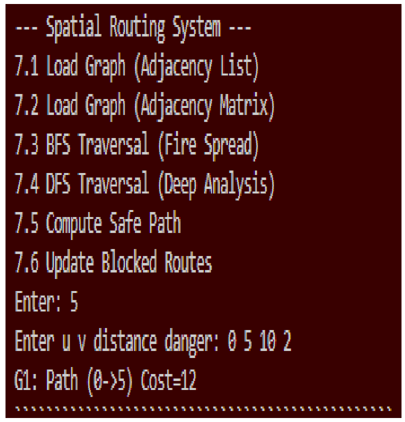
Functions:

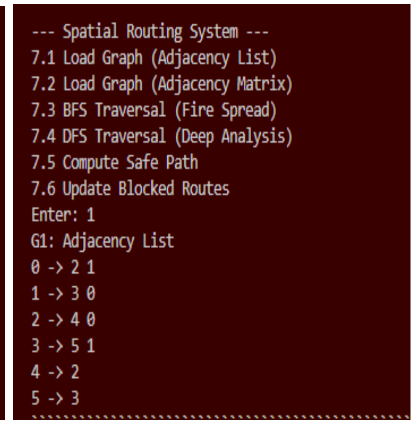
- BFS traversal
- DFS traversal
- Safe path calculation



```
IFAMDS MAIN MENU
1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 7

--- Spatial Routing System ---
7.1 Load Graph (Adjacency List)
7.2 Load Graph (Adjacency Matrix)
7.3 BFS Traversal (Fire Spread)
7.4 DFS Traversal (Deep Analysis)
7.5 Compute Safe Path
7.6 Update Blocked Routes
Enter: 3
Enter start node (0-5): 0
G1 G1 BFS: 0 2 1 4 3 5
G BFS: 0 1 2 3 4 5
```


```
--- Spatial Routing System ---
7.1 Load Graph (Adjacency List)
7.2 Load Graph (Adjacency Matrix)
7.3 BFS Traversal (Fire Spread)
7.4 DFS Traversal (Deep Analysis)
7.5 Compute Safe Path
7.6 Update Blocked Routes
Enter: 4
Enter start node (0-5): 0
G1 DFS: 0 2 4 1 3 5
G DFS: 0 1 3 5 2 4
```


```
--- Spatial Routing System ---
7.1 Load Graph (Adjacency List)
7.2 Load Graph (Adjacency Matrix)
7.3 BFS Traversal (Fire Spread)
7.4 DFS Traversal (Deep Analysis)
7.5 Compute Safe Path
7.6 Update Blocked Routes
Enter: 5
Enter u v distance danger: 0 5 10 2
G1: Path (0->5) Cost=12
```

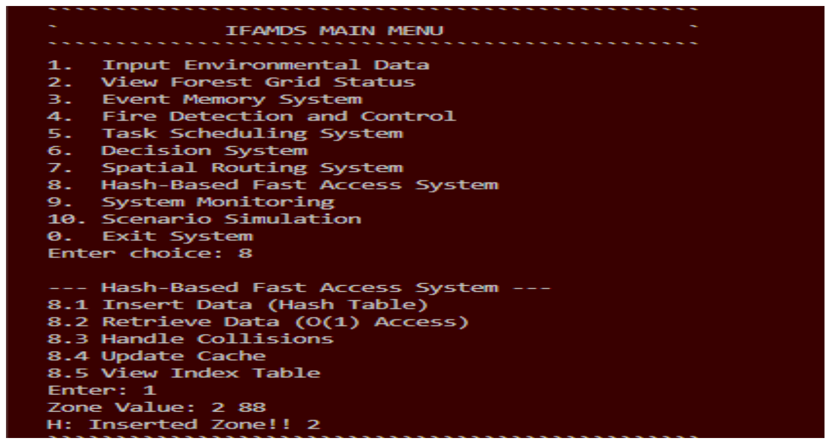

```
--- Spatial Routing System ---
7.1 Load Graph (Adjacency List)
7.2 Load Graph (Adjacency Matrix)
7.3 BFS Traversal (Fire Spread)
7.4 DFS Traversal (Deep Analysis)
7.5 Compute Safe Path
7.6 Update Blocked Routes
Enter: 1
G1: Adjacency List
0 -> 2 1
1 -> 3 0
2 -> 4 0
3 -> 5 1
4 -> 2
5 -> 3
```

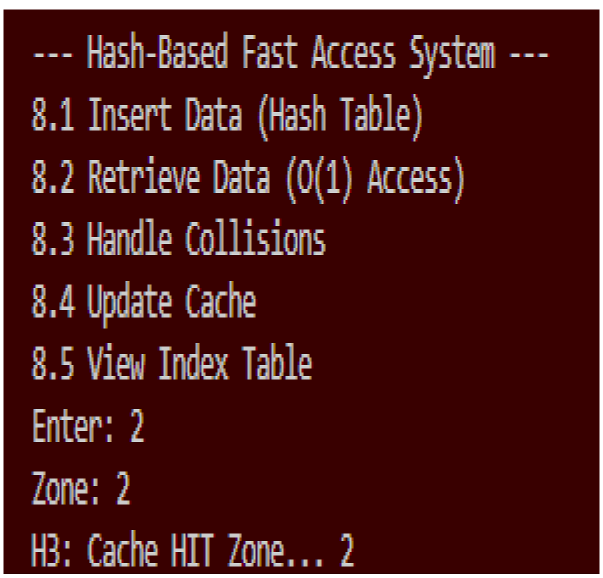
3.8 Hash-Based Access System

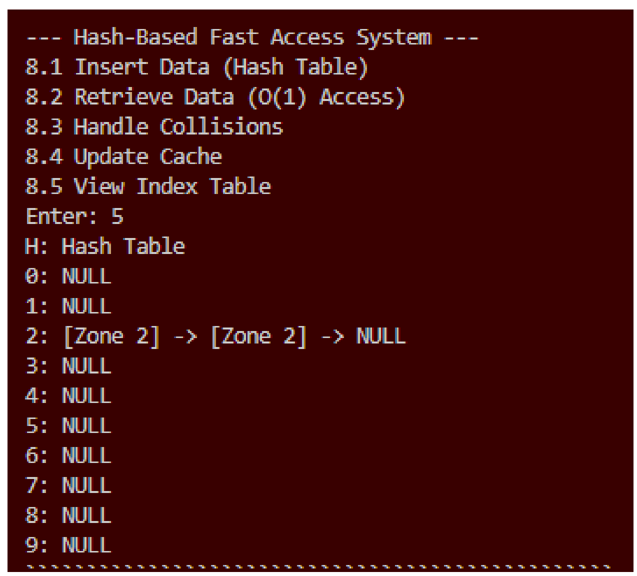
This module provides fast data access using hash tables.

Functions:

- Insert records
- Search records
- Update cache


```
IFAMDS MAIN MENU
1.  Input Environmental Data
2.  View Forest Grid Status
3.  Event Memory System
4.  Fire Detection and Control
5.  Task Scheduling System
6.  Decision System
7.  Spatial Routing System
8.  Hash-Based Fast Access System
9.  System Monitoring
10. Scenario Simulation
0.  Exit System
Enter choice: 8

--- Hash-Based Fast Access System ---
8.1 Insert Data (Hash Table)
8.2 Retrieve Data (O(1) Access)
8.3 Handle Collisions
8.4 Update Cache
8.5 View Index Table
Enter: 1
Zone Value: 2 88
H: Inserted Zone!! 2
```


```
--- Hash-Based Fast Access System ---
8.1 Insert Data (Hash Table)
8.2 Retrieve Data (O(1) Access)
8.3 Handle Collisions
8.4 Update Cache
8.5 View Index Table
Enter: 2
Zone: 2
H3: Cache HIT Zone... 2
```


```
--- Hash-Based Fast Access System ---
8.1 Insert Data (Hash Table)
8.2 Retrieve Data (O(1) Access)
8.3 Handle Collisions
8.4 Update Cache
8.5 View Index Table
Enter: 5
H: Hash Table
0: NULL
1: NULL
2: [Zone 2] -> [Zone 2] -> NULL
3: NULL
4: NULL
5: NULL
6: NULL
7: NULL
8: NULL
9: NULL
```

3.9 System Monitoring

This module monitors system load and performance.

Functions:

- Analyze queue load
- Track execution time
- Detect bottlenecks

```
--- System Monitoring ---
9.1 Monitor System Load
9.2 Track Execution Time
9.3 Detect Bottlenecks
9.4 Optimize Performance
9.5 View System Health
Enter: 1
Queue number (1-4) and task count: 3 25
SYS: Load Analysis:
Q1-Routine: 0 tasks | Load=0 [OK]
Q2-Surveillance: 0 tasks | Load=0 [OK]
Q3-Emergency: 25 tasks | Load=0.5 [OK]
Q4-Decision: 0 tasks | Load=0 [OK]
```

```
--- System Monitoring ---
9.1 Monitor System Load
9.2 Track Execution Time
9.3 Detect Bottlenecks
9.4 Optimize Performance
9.5 View System Health
Enter: 2
Latency: 0 ms
```

```
--- System Monitoring ---
9.1 Monitor System Load
9.2 Track Execution Time
9.3 Detect Bottlenecks
9.4 Optimize Performance
9.5 View System Health
Enter: 5
SYS: System Health:
Latency: 0 ms
Bottleneck: None
SYS: Load Analysis:
Q1-Routine: 0 tasks | Load=0 [OK]
Q2-Surveillance: 0 tasks | Load=0 [OK]
Q3-Emergency: 25 tasks | Load=0.5 [OK]
Q4-Decision: 0 tasks | Load=0 [OK]
Status: HEALTHY
```

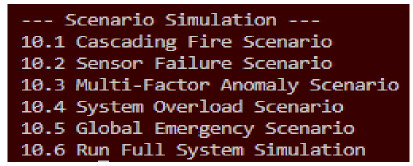
```
--- System Monitoring ---
9.1 Monitor System Load
9.2 Track Execution Time
9.3 Detect Bottlenecks
9.4 Optimize Performance
9.5 View System Health
Enter: 4
SYS: Optimizing...
SYS: Optimization complete.
```

3.10 Scenario Simulation

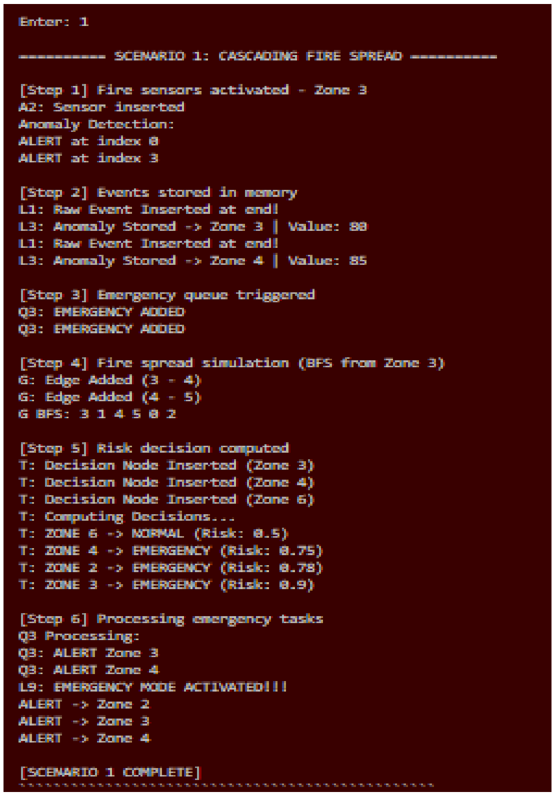
This module demonstrates different emergency situations.

Scenarios:

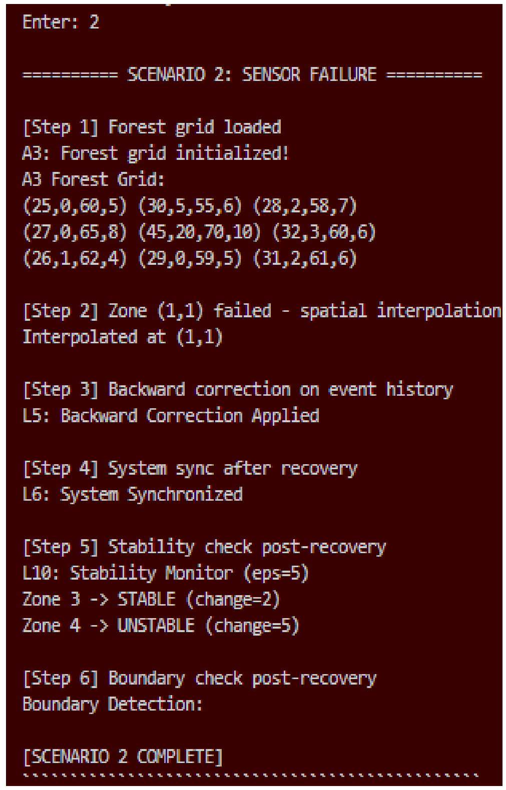
- Cascading fire
- Sensor failure
- Multi-factor anomaly
- System overload
- Global emergency



```
--- Scenario Simulation ---
10.1 Cascading Fire Scenario
10.2 Sensor Failure Scenario
10.3 Multi-Factor Anomaly Scenario
10.4 System Overload Scenario
10.5 Global Emergency Scenario
10.6 Run Full System Simulation
```



```
Enter: 1

========== SCENARIO 1: CASCADING FIRE SPREAD ==========

[Step 1] Fire sensors activated - Zone 3
A2: Sensor inserted
Anomaly Detection:
ALERT at index 0
ALERT at index 3

[Step 2] Events stored in memory
L1: Raw Event Inserted at end!
L3: Anomaly Stored -> Zone 3 | Value: 88
L1: Raw Event Inserted at end!
L3: Anomaly Stored -> Zone 4 | Value: 85

[Step 3] Emergency queue triggered
Q3: EMERGENCY ADDED
Q3: EMERGENCY ADDED

[Step 4] Fire spread simulation (BFS from Zone 3)
G: Edge Added (3 - 4)
G: Edge Added (4 - 5)
G BFS: 3 1 4 5 0 2

[Step 5] Risk decision computed
T: Decision Node Inserted (Zone 3)
T: Decision Node Inserted (Zone 4)
T: Decision Node Inserted (Zone 6)
T: Computing Decisions...
T: ZONE 6 -> NORMAL (Risk: 0.5)
T: ZONE 4 -> EMERGENCY (Risk: 0.75)
T: ZONE 2 -> EMERGENCY (Risk: 0.78)
T: ZONE 3 -> EMERGENCY (Risk: 0.9)

[Step 6] Processing emergency tasks
Q3 Processing:
Q3: ALERT Zone 3
Q3: ALERT Zone 4
L9: EMERGENCY MODE ACTIVATED!!!
ALERT -> Zone 2
ALERT -> Zone 3
ALERT -> Zone 4

[SCENARIO 1 COMPLETE]
```



```
Enter: 2

========== SCENARIO 2: SENSOR FAILURE ==========

[Step 1] Forest grid loaded
A3: Forest grid initialized!
A3 Forest Grid:
(25,0,60,5) (30,5,55,6) (28,2,58,7)
(27,0,65,8) (45,20,70,10) (32,3,60,6)
(26,1,62,4) (29,0,59,5) (31,2,61,6)

[Step 2] Zone (1,1) failed - spatial interpolation
Interpolated at (1,1)

[Step 3] Backward correction on event history
L5: Backward Correction Applied

[Step 4] System sync after recovery
L6: System Synchronized

[Step 5] Stability check post-recovery
L10: Stability Monitor (eps=5)
Zone 3 -> STABLE (change=2)
Zone 4 -> UNSTABLE (change=5)

[Step 6] Boundary check post-recovery
Boundary Detection:

[SCENARIO 2 COMPLETE]
```

```
========== SCENARIO 3: MULTI-FACTOR ANOMALY ==========

[Step 1] Multiple anomaly signals detected
A2: Sensor inserted
A2: Sensor inserted
Anomaly Detection:
ALERT at index 0
ALERT at index 3
ALERT at index 4
ALERT at index 5

[Step 2] Anomalies stored in L3 stream
L3: Anomaly Stored -> Zone 5 | Value: 60
L3: Anomaly Stored -> Zone 6 | Value: 55

[Step 3] Fire classification computed (T7)
T7: Fire Zone 5 Level=0.75
T7: Fire Zone 6 Level=0.65
T7: Fire Classification:
Zone 6 FireLevel=0.65 [MODERATE]
Zone 5 FireLevel=0.75 [MAJOR FIRE]

[Step 4] Wildlife activity detected (T8)
T8: Wildlife Zone 5 Activity=0.9
T8: Wildlife Zone 6 Activity=0.8
T8: Wildlife Activity:
Zone 6 Activity=0.8 [FLEEING - FIRE LIKELY]
Zone 5 Activity=0.9 [FLEEING - FIRE LIKELY]

[Step 5] Human intrusion detected (T9)
T9: Human Zone 5 Risk=0.63
T9: Human Zone 6 Risk=0.48
T9: Human Activity:
Zone 6 HumanRisk=0.48
Zone 5 HumanRisk=0.63 [INTRUSION DETECTED]

[Step 6] All anomalies escalated to emergency queue
PROMOTED TO EMERGENCY QUEUE
PROMOTED TO EMERGENCY QUEUE
Q2: Enqueued Surveillance Task!
Q2: Enqueued Surveillance Task!

[Step 7] Processing all queues
Q3 Processing:
Q3: ALERT Zone 5
Q3: ALERT Zone 6
Q2 Processing:
Q2: Dequeued Zone 7
Q2: Dequeued Zone 8

[SCENARIO 3 COMPLETE]
``````````````````````````````````````````````
```

```
========== SCENARIO 4: SYSTEM OVERLOAD ==========

[Step 1] High traffic flooding Q2
Q2: Enqueued Surveillance Task!
Q2: Enqueued Surveillance Task!
Q2: Enqueued Surveillance Task!
Q2: Enqueued Surveillance Task!
Q2: Enqueued Surveillance Task!

[Step 2] Monitor analyzing load
SYS: Load Analysis:
Q1-Routine: 1 tasks | Load=0.02 [OK]
Q2-Surveillance: 5 tasks | Load=0.1 [OK]
Q3-Emergency: 25 tasks | Load=0.5 [OK]
Q4-Decision: 0 tasks | Load=0 [OK]
SYS: No bottleneck.

[Step 3] Pausing low-priority queues
Q1: Paused
Q4: Paused
Queue Status:
Q1 (Routine):   PAUSED
Q2 (Surveillance): RUNNING
Q3 (Emergency):   RUNNING
Q4 (Decision):   PAUSED

[Step 4] Rebalancing Q2 -> Q1
Balanced: Moved Q2 -> Q1
Balanced: Moved Q2 -> Q1

[Step 5] Resuming system
Q1: Resumed
Q4: Resumed
Q1 Processing:
Q1: Dequeued Zone 1
Q1: Dequeued Zone 2
Q2 Processing:
Q2: Dequeued Zone 3
Q2: Dequeued Zone 4
Q2: Dequeued Zone 5

[Step 6] Performance optimization
SYS: Optimizing...
SYS: Optimization complete.
SYS: System Health:
Latency: 2 ms
Bottleneck: None
SYS: Load Analysis:
Q1-Routine: 1 tasks | Load=0.02 [OK]
Q2-Surveillance: 5 tasks | Load=0.1 [OK]
Q3-Emergency: 25 tasks | Load=0.5 [OK]
Q4-Decision: 0 tasks | Load=0 [OK]
Status: HEALTHY

[SCENARIO 4 COMPLETE]
```
```

```
========== SCENARIO 5: GLOBAL EMERGENCY ==========

[Step 1] Multi-zone emergency signals received
Q3: EMERGENCY ADDED
Q3: EMERGENCY ADDED
Q3: EMERGENCY ADDED
Q3: EMERGENCY ADDED

[Step 2] Global risk tree updated
T12: Global Node Zone 1
T12: Global Node Zone 2
T12: Global Node Zone 3
T12: Global Node Zone 4
T12: Global Emergency Tree:
Zone 2 Risk=0.78
Zone 3 Risk=0.88
Zone 2 Risk=0.9
Zone 4 Risk=0.92
Zone 1 Risk=0.95

[Step 3] Global alert check
T12: Total Risk = 4.43
T12: GLOBAL EMERGENCY ACTIVATED!

[Step 4] Deep path analysis (DFS)
G DFS: 0 1 3 4 2 5

[Step 5] Zone data indexed in hash table
H: Inserted Zone!! 1
H: Inserted Zone!! 2
H: Inserted Zone!! 3
H: Inserted Zone!! 4
H3: Cached Zone!! 1
H3: Cache updated Zone 2
H3: Cache HIT Zone... 1
H3: Cache HIT Zone... 2
H2: Collision Slots:
Index 2: [Zone 2] -> [Zone 2] -> [Zone 2] -> NULL

[Step 6] Emergency response dispatch
Q3 Processing:
Q3: ALERT Zone 1
Q3: ALERT Zone 2
Q3: ALERT Zone 3
Q3: ALERT Zone 4

[SCENARIO 5 COMPLETE]
```

4. Scenario Explanation

Scenario 1: Cascading Fire

A fire starts in Zone 3 and spreads toward nearby zones. Sensor readings are collected and checked against baseline values. Emergency events are sent to the priority queue, while BFS is used to simulate fire spread across the graph.

Scenario 2: Sensor Failure

Sensors in Zone 2 start sending incorrect data. The system detects the problem, restores the last stable state, and rebuilds missing values using nearby sensor information.

Scenario 3: Multi-Factor Anomaly

The system detects multiple dangers at the same time, including fire risk, animal movement, and human intrusion. A combined risk score is calculated, and high-risk zones are moved to the emergency queue.

Scenario 4: System Overload

A large number of sensor updates overloads the system queues. Critical tasks are moved to the emergency queue while low-priority tasks are paused until the system becomes stable again.

Scenario 5: Global Emergency

A large-scale emergency affects multiple forest zones at the same time. The system checks for conflicting data, restores the correct global state, and issues a synchronized emergency response.

5. Conclusion

The IFAMDS project successfully demonstrates the use of multiple data structures and algorithms in a real-world forest monitoring simulation. The system can monitor environmental conditions, detect emergencies, process tasks, perform routing operations, and manage different scenarios efficiently. This project helped improve understanding of arrays, linked lists, queues, trees, graphs, hashing, BFS, and DFS in practical applications.